Assignment – 2.4

Network Threats and Mitigation & Physical Security and Risk

Q - 1. Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

- **Common Network Threats with Real-World Examples:**

No.	Network Threat	Simple Meaning	Real-Life Example
1	**Phishing**	Fake messages or links used to steal information.	**WhatsApp:** Someone sends a fake prize-winning link to steal your details.
2	**DDoS Attack**	Sending too much traffic to make a website stop working.	**Instagram:** Too many fake requests can make Instagram unavailable.

No.	Network Threat	Simple Meaning	Real-Life Example
3	**Man-in-the-Middle**	A hacker secretly reads your online communication.	**Paytm:** A hacker tries to steal information through an unsecured public Wi-Fi network.
4	**Malware**	Harmful software that can damage your phone or steal data.	**WhatsApp:** Downloading a virus-infected file can harm your phone.
5	**Password Attack**	Trying to steal or guess someone's password.	**Instagram:** A hacker tries to guess your Instagram password to access your account.

Q - 2. Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value. Constraint: The script should print the number of attempts it took to guess the correct PIN.

- **Brute-Force Attack Using Python:**

```
correct_pin = "1234"
attempts = 0

for i in range(10000):
    guess = str(i).zfill(4)
    attempts += 1

    if guess == correct_pin:
        print("PIN found:", guess)
        print("Total attempts:", attempts)
        break
```

- **Output:**

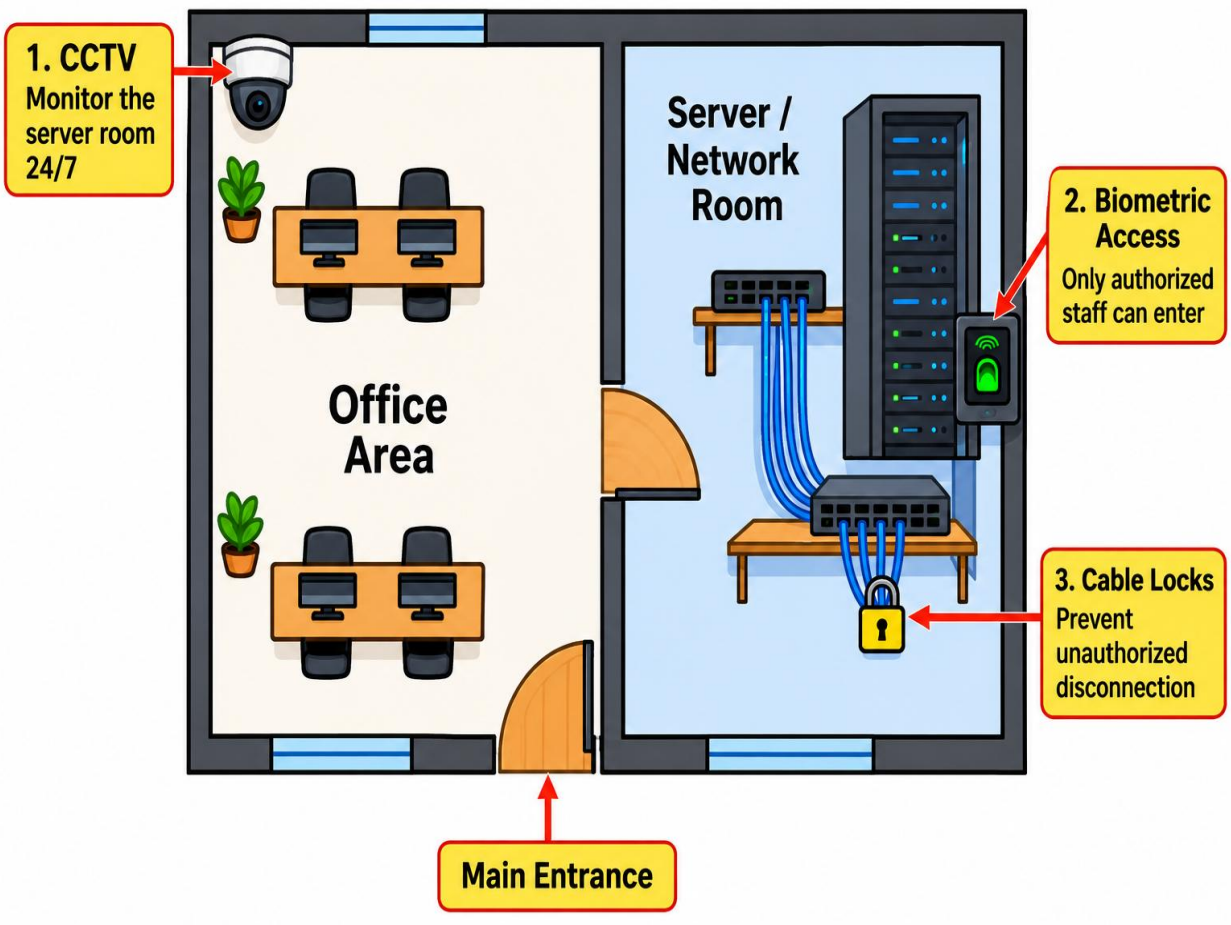
```
PIN found: 1234
Total attempts: 1235
```

Q - 3. List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

➢ **Physical Security Risks in a Server Room:**

No.	Risk	Solution
1.	**Fire:** Can damage servers.	Install fire alarms.
2.	**Unauthorized Access:** Unknown people can enter.	Use ID cards or fingerprint locks.
3.	**Power Failure:** Servers may stop working.	Use a UPS and generator.
4.	**Water Leakage:** Can damage computers.	Install water leak detectors.

Q - 4. Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).



Q - 5. Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

- **Network Security Checklist for a College Computer Lab:**

1. **Strong Passwords:** Use strong passwords on all lab computers and Wi-Fi.
2. **Install Antivirus:** Install antivirus software to protect computers from viruses.
3. **Use Firewall:** Use a firewall to block hackers and unwanted access.
4. **CCTV Cameras:** Install CCTV cameras to monitor the computer lab.
5. **Restricted Access:** Allow only authorized people to enter the server room.
6. **Data Backup:** Take regular backups of important college data.
7. **Secure Devices:** Keep routers, switches, and servers in locked cabinets.